

Your principal reduction target is too high.

When the loan term is specified, the principal reduction target must not be so high as to bring the loan balance below zero before the loan is completed.

Your printer code string is not properly formatted.

The two formats that are supported by *Per%Sense* are explained with examples on page 38.

Glossary

Financial Terms and Per%Sense Words

Actuarial table

A table of probabilities that a person alive today will still be alive *N* years in the future. Actuarial tables always take account of age and, generally, sex. They may also be more specific; e.g. for white, male non-smokers with family income between \$50,000 and \$99,999.

Actuarial rule

The standard method of computing interest on a loan. Each month (or interest period), the amount of the outstanding balance is multiplied by the *loan rate* and the result is the amount of interest due. If the monthly loan payment is greater than the computed interest, then the difference is applied to reduce the outstanding balance (principal). If the payment is less than the interest, the difference is tacked on to the outstanding balance. The other "rule," used less frequently, is the *US rule*. Loans computed with the Actuarial rule are sometimes called **direct reduction loans**.

Advance/Arrears

Loan payments may be scheduled so that each month's payment includes interest for the month just ended (interest in arrears) or interest for the month to come (interest in advance). The former is more common.

Amortization schedule (or table)

A table listing each of the payments made on a loan, and showing how much of that payment is devoted to interest and to principal.

Amount borrowed

In the *Per%Sense Mortgage* screen, the amount borrowed refers to the price of the house less the down payment.

Amount financed

The actual proceeds from a loan. Often a borrower is obliged to pay *points* and *prepaid interest* for the first irregular interest period on the day a loan settles. The amount financed is the loan amount minus these charges.

APR (Annual Percentage Rate)

The interest rate that every lender is obligated by *The Truth in Lending Act* to reveal in its statement to the borrower. It is defined as that rate which makes the sum of the values of all payments on a loan equal to the amount financed as of the loan date. It is always quoted as a *loan rate*. When no points are charged, the APR is the same as the quoted *loan rate*; with points, the APR is higher than the *loan rate*.

Backward computation

A computation starting with a given value and interest rate, whose target is the amount of any one payment, or a periodic payment, or a date in the payment stream specified at the top of the **Present Value** screen. Most commonly, the **Present Value** screen is used with input on the top and output on the bottom. In a "backward" computation, the bottom line is fully specified and one cell at top is left blank, to be computed.

Balloon payment

Any lump sum deposit used in combination with *periodic payments* to repay a loan. Often a balloon payment occurs at the end of a loan, but multiple balloon payments may be scheduled at any time.

Basis days

Per%Sense offers the option of using the 360-day calendar or the 365-day calendar. The choice is made in the **Computational Settings** screen (press (Alt)-(S)). This determines how the time interval between any two dates is computed. It affects the values of **Single Payments** in the **Present Value** screen, and it affects how interest is computed on deposits in the **Chronological** screen. With 365 selected, the time interval between two dates in the **Chronological** screen and the **Present Value** screen is taken as the exact number of intervening days divided by 365.25. In the **Amortization** screen, with 365 selected, interest on partial interest periods is computed as the number of days divided by 365, in agreement with most banks.

The number of days between two dates computed according to the standard 30/360 calendar begins with a simple idea, but becomes

Basis point

painfully technical for exceptional cases. The number of days between dates $m_1/d_1/y_1$ and $m_2/d_2/y_2$ is taken to be

$$360 \times (y_2 - y_1) + 30 \times (m_2 - m_1) + (d_2 - d_1) + \text{correction}$$

The "correction" term is the part of the standard that subsumes all the complexity.

One hundredth of one percent (0.01%).

Bond

A loan instrument, usually including periodic interest payments, and culminating in a full repayment of principal on the bond's date of maturity. American corporate bonds are commonly issued in denominations of \$1,000 and make two interest payments per year until maturity.

Canadian rate

It is conventional in Canadian mortgages to quote interest as if it were compounded semi-annually, even if the actual payments occur monthly. The Canadian rate, like *true rate*, *loan rate*, and *yield* is one way of measuring or specifying the interest rate.

Cash flow

Any set of payments and dates, in the past or the future or both.

Cash required

The amount of money the buyer of a house needs at settlement. In the *Per%Sense Mortgage* screen, this column holds the sum of the down payment amount and the bank's points. In real life, there are other small miscellaneous fees that add to the cash required to settle on a house.

COLA (Cost of Living Adjustment)

Annuities and structured legal settlements sometimes include provisions for periodic payments that increase from year to year to compensate for inflation. COLA percentage is usually quoted as a *yield*, meaning that it is not intended to be compounded within a one-year period, or that it is "pre-compounded."

Compound interest

In compound interest calculations, interest is computed as a percentage of the current balance, regardless of how much of that balance is principal and how much is interest. Mathematically, the most fundamental way of computing interest is continuous compounding, but the arithmetic is easier for periodic compounding.

Continuous compounding

The most logical and standard way of computing interest leads, unfortunately, to some mathematical sophistication. Interest is computed on a current balance, which is a moving target consisting of principal plus interest already accrued.

For logical consistency, this should be considered a continuous process. The interest rate becomes an exponential growth rate.

But before the time of electronic calculators, the computation of exponential growth was a complicated and inconvenient process, so the custom arose of compounding interest at periodic intervals. This is still standard practice in loan computations.

Simple interest is computed within a compounding period. At the end of the period, the interest amount is tacked on to principal, and simple interest can again be computed for the next period. This leads to an approximation of exponential growth, and can be computed without an electronic calculator's exponential function.

The difference between continuous compounding and periodic compounding can be exactly compensated by using a *loan rate* instead of a *true rate*.

Coupon

The total yearly interest paid by a bond. It is quoted as a percentage. Most American corporate bonds make two interest payments per year, and each one is half the coupon times the price of the bond.

Daily rate

A "true" interest rate compounded over just one day. Because one day is such a short time, the daily rate is numerically very close to the true rate. It is appropriate to use the daily rate when computing bank interest on a passbook account with "daily compounding." Occasionally, a "365-day" loan is also computed based on daily interest.

Effective annual rate

and Yield.

Frontward computation

A computation of the present value of a cash flow. The cash flow is specified at the top of the **Present Value** screen, and the value is computed at bottom. Computations of IRR (internal rate of return) are also considered frontward, in that they summarize a stream of payments in a single number.

Future value

The future value of a cash flow is the balance at a specified date of a hypothetical interest-bearing account, in which that cash flow is deposited. Future value is a function of the cash flow, the future date, and the interest rate or rates.

Graduated payments

Sometimes a loan repayment is structured with smaller payments in the early years and larger payments as time goes on.

Internal Rate of Return (IRR)

The true measure of investment performance over time. IRR is the same as time-weighted average return, and is closely related to the concepts APR and bond yield.

IRR is a single interest rate that summarizes all investment monies and all returns on that investment.

To set up an IRR calculation in the **Present Value** Screen:

- 1- Enter the date and amount of the original investment in the bottom block of the screen, labeled "Present Value". Leave the three rate and yield cells blank.
- 2- List all payments that you receive in return for the investment at the top of the screen - periodic payments on the right and one-time payments on the left.
- 3- If additional monies are invested after the original investment date, enter the additional investments as negative payments alongside the positive amounts that are returned to you.
- 4- Press **(Enter)** and read the IRR from the True Rate column at the bottom of the screen.

Loan rate

A measure of interest percentage. The *true rate* compounded over one interest period. The quoted *loan rate* actually represents a smaller true interest rate if payments are less frequent, and a larger number if payments are more frequent. For monthly payments, the *loan rate* is a little larger than the *true rate* and somewhat smaller than the *yield*.

Monthly total

The total amount of monthly mortgage payments. In many areas, it is conventional for the bank to collect real estate tax and insurance premiums from the borrower, and to include these in the monthly mortgage bill.

Negative amortization

If deposits in the early phase of a loan repayment are insufficient to cover the interest that is accruing, the principal will, for a time, increase from payment to payment. (Later in the loan schedule, payments must increase, or balloon payments must be added to assure that the loan is paid off.)

Percent down

Percentage of the purchase price of a house that is advanced as a down payment. This portion is not borrowed, so no points are charged on the down payment.

Periodic compounding

It is conventional in most loans and savings situations to compute simple interest for a short period of time, then add the interest amount to the principal amount. During the next period, interest is computed on the sum of the principal and the interest added from the last period. The result is an approximation to continuous compounding. If a substitution of the *loan rate* for the *true rate* is made, periodic compounding becomes exactly equivalent to continuous compounding.

For loans that are repaid monthly, the compounding period is almost always taken to be one month. For Canadian mortgages, the period is taken as half a year, regardless of the payment frequency. For corporate and municipal bonds, the period is usually half a year.

Periodic payment

Any deposit made at regular intervals into a savings account or in repayment of a loan.

Points

Banks often charge an up-front fee, quoted as a percentage of the loan amount, in addition to periodic interest on a loan. A "point" is one percent of the amount of the loan, and is usually deducted directly from the amount of the loan, and paid at settlement.

Prepaid interest

If a loan is settled on a different day of the month than the day that payments will be due, there are two ways the situation is commonly handled. The loan payments can be adjusted so that they are all equal and reflect the partial month's interest. Or the first payment may be scheduled more than one month in advance, and interest for the extra partial month is paid at settlement from the proceeds of the loan. This latter arrangement is referred to as prepaid interest.

Present Value

The present value of a cash flow from the past is the balance at present of a hypothetical interest-bearing account in which that cash flow would have been deposited. Present value is a function of the cash flow, the date, and the interest rate or rates. Present value of a future cash flow is the balance of a hypothetical interest-bearing account from which that cash flow may be withdrawn, emptying the account with the last withdrawal. You might also think of the present value of a future cash flow as the amount of a loan that could be repaid by that cash flow.

Principal moratorium (Interest-only payments)

Clause in a loan agreement that specifies that deposits during the first phase of loan repayment are just enough to cover interest, and the principal balance remains constant. (Later in the loan schedule, payments must increase, or *balloon payments* must be added to assure that the loan is paid off.)

Rule of 78

The *Rule of 78* is an accounting fiction that divides loan payments between interest and principal in a way that favors the lender in case of early repayment. The computation of payment amount is not affected.

If a loan is paid in fixed, constant installments, interest is concentrated in the early payments and principal in the later payments – this much is just mathematics. But the *Rule of 78* permits even more of the early payments to be considered interest than would be true under a fixed-rate computation. Lenders like this because it penalizes borrowers for paying off their loans early. At one time, the *Rule of 78* was also an accepted method for calculating tax liability, and it permitted the borrower to deduct more interest sooner. Although it is no longer acceptable for tax calculations, some leases and consumer-lending contracts continue to specify a *Rule of 78* computation of the amount owed if the borrower decides to repay the loan ahead of schedule.

A *Rule of 78* computation begins with a standard amortization of interest and principal, then takes the total interest and re-allocates

Settlement

it over the repayment period. The curious name derives from the fact that for a one-year loan with 12 monthly installments, $\frac{12}{78}$ of the interest is charged in the first month, $\frac{11}{78}$ in the second month, etc. 78 is the sum of the integers from 1 to 12.

The meeting at which papers are exchanged and a loan is delivered to the borrower. Interest accrues from the day of settlement. Often, the borrower must pay up-front fees out of the loan he has just received, including *prepaid interest* and *points*.

Simple interest

A method of computing interest whereby the interest rate is applied only to the principal amount, and no interest is computed on the interest itself.

The concept of simple interest arose from a desire to avoid the mathematics of exponentials, which is essential to compound interest computations. However, there is no clear conceptual foundation for simple interest, and in practice, all schemes for computing simple interest have become a patchwork of inelegant rules.

The concept of simple interest becomes ambiguous when payments are made on a loan balance: Are those payments to be credited against interest or against principal?

For standard loans, repaid in equal, periodic installments, the issue of simple vs. compound interest never arises: the interest is being paid currently, so there is never an accrued interest balance on which to compute compound interest.

For non-standard loan repayments involving negative amortization, the concept of simple interest is formalized in the *US rule*.

Simple interest and civil law

Simple interest is mandated by some states for court-awarded claims for past damages. Simple interest on past payments has the effect of reducing the present value. However, simple interest used to compute the value of future payments results in an increased present value. So the latter is never done in practice.

If past damages were mitigated by income or partial payments to the plaintiff, computation of simple interest again becomes ambiguous, and there are no clear rules for how the computation is to be done.

Targeted principal reduction

If a loan is paid in fixed monthly installments, the part of each payment that is applied to principal increases through the life of the loan. If the loan is scheduled with targeted principal reductions, then each payment includes all interest for that period plus a constant dollar repayment of principal. The payment amounts actually decrease from one period to the next.

Term life insurance

A whole life insurance policy promises to pay a benefit no matter when the insured dies. Term life insurance, on the other hand, pays only if the insured dies before a given date. *Per%Sense* permits modeling of term life insurance with the *if before* date, at the bottom right of the **Actuarial Data** window.

True rate

The most fundamental way to measure an interest rate. No compounding is included. The *true rate* is almost exactly equal to the *daily rate*, and is a little less than the *loan rate* and the *yield*.

US rule

This rule (which is actually mandated in just a few states) requires that no interest on interest be charged. In standard periodic loan schedules, each monthly payment is sufficient to cover interest for that period and make an additional contribution to principal. In this case, the *US rule* is moot. But in *negative amortization*, or circumstances where interest is accruing faster than the payments are coming in, the *US rule* mandates that the shortfall be treated separately, and two balances are maintained through the life of the loan, for interest and principal. Interest is computed only on the latter, and shortfalls accumulate in the former. Whenever a payment is made, it is applied first against the interest balance, and any surplus is applied against the principal balance.

Sometimes people speak of simple interest in the repayment of a loan. This concept is problematic, and needs further definition before it can be considered an accounting rule. There must be "two kinds of money," that which earns interest and that which does not, and a rule for saying into which class each dollar falls. The *US rule* is really a formalization of the concept of simple interest.

The opposite of the *US rule* is the **Actuarial Rule**, or *direct reduction loan*.

Variable rate

An interest rate that changes over time, either in the repayment of a loan, or in a savings situation.

Variable rate loan

A loan that is written at a current interest rate, but with a proviso that the rate may rise or fall in the future with the movement of some agreed-on rate that is set by a third party.

Whole life insurance

A whole life insurance policy promises to pay a benefit no matter when the insured dies. Term life insurance, on the other hands, pays only if the insured dies before a given date.

Yield

A measure of interest percentage. The *true rate* compounded over one full year. An account with 100 dollars and a *yield* of $X\%$ will hold $100 + X$ dollars a full year later (after all compounding). (*Effective annual rate* has the same meaning.)

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